

# Evolution of the Vocabulary of a Text and Distribution of Occurrences

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## 1 Introduction

The objective of this report is to conduct a computational investigation on a corpus of Italian literature texts and a Chinese literary work (*Romance of the Three Kingdoms*). It will analyze the relationship between the length of a text and the variety of the vocabulary used, as well as the frequency distribution of word usage, considering texts from different languages to evaluate the preservation of any statistical properties.

## 2 Results

### 2.1 a) Vocabulary growth dynamics

The analysis answers the question of how the number of distinct words ( $V$ ) increases with the total text length ( $N$ ). As highlighted in **Figure 1 (left)**, the growth seems to follow a power law. The cross-linguistic comparison reveals a structural divergence: while Italian texts show steady growth due to morphological inflection, the Chinese text (black line, analyzed by individual ideograms) exhibits much earlier saturation. As shown in **Figure 1 (right)**, the data point for the Chinese text is positioned significantly lower than the Italian trend despite the vast quantity of ideograms. This indicates that the repertoire of graphic symbols in Chinese is a more limited set that tends to be exhausted quickly, unlike the vocabulary of European languages.

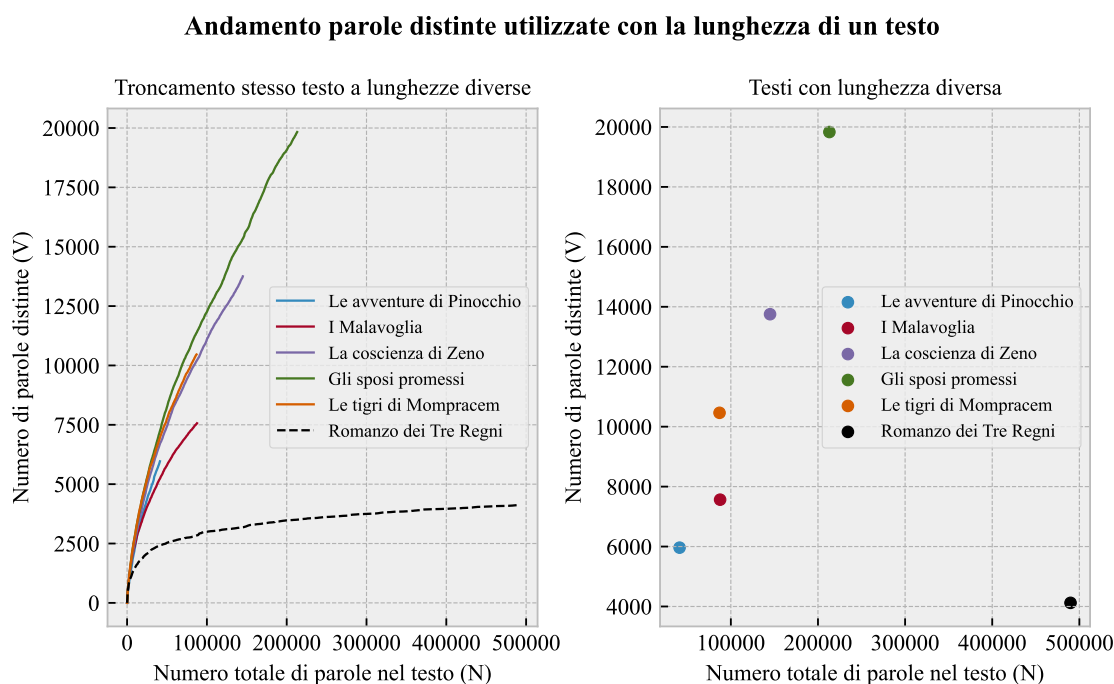


Figure 1: *Left*: vocabulary growth. The black line (Chinese) shows the lower richness of available ideograms. *Right*: comparison of total lexical density.

## 2.2 b) Occurrence distribution

The occurrence histogram analyzed in **Figure 2** shows a distribution that follows a constant slope both for truncations of the same text (**Figure 2, left**) and for the comparison between different works (**Figure 2, right**). The inclusion of the Chinese text (**Figure 2, right, black line**) shows that despite its ideographic nature and the different structure of the language, the frequency distribution of ideograms almost perfectly follows the slope of the Italian texts. The "zig-zag" pattern in the tail of the distribution is a statistical phenomenon due to the discreteness of the data for very high-frequency terms, but it does not invalidate the power law that governs the human linguistic system regardless of the language considered.

### Comportamento dell'istogramma di occorrenza delle parole

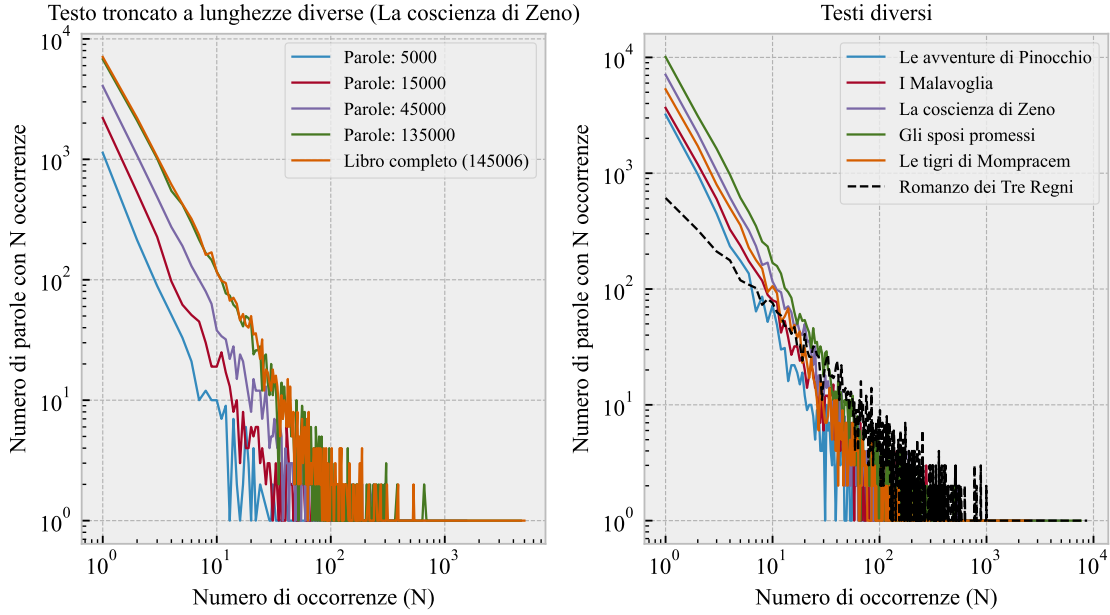


Figure 2: *Left*: slope stability for truncations of a single text.

*Right*: the alignment of the black line confirms the universality of the law for the Chinese language as well.

## 3 Conclusions

The evidence shows that language structure is characterized by power laws and, although lexical richness and the speed of vocabulary saturation depend heavily on the writing system and the author's style, the distribution of usage frequencies emerges as a universal property of human communication.

## 4 Methods

The dataset was built via scraping (for Italian texts) and manual acquisition (for the Chinese text) from *Wikisource*. Data cleaning in Python involved two different paths to handle the two languages:

- **Italian Texts:** Pre-cleaning was performed during the scraping phase, followed by post-processing aimed at separating terms using spaces and punctuation. For the handling of apostrophes (e.g., "*l'orario*"), it was decided to separate them into distinct words ("*l*", "*orario*").
- **Chinese Text:** Given the absence of graphic spacers, individual Hanzi characters were extracted using the Unicode property `IsHan` via the `regex` module. Each ideogram was treated as a minimum lexical unit (each ideogram counts as a "word").

The resulting data was structured in a *pandas* DataFrame to calculate the occurrences and cumulative frequencies required to generate the plots.